23/7/2020

1. Study of various Network Commands used in Linux and Windows

AIM :-

Study of various networking commands used in Linux and Windows.

Commands :-

i). arp -a :- Address resolution protocal is used to resolve networks IP and MAC address of your router by sending IP packets.

```
[Hostname]$ arp -a
```

Output :-

? (172.16.10.82) at d8:bb:c1:c5:cd:4f [ether] on enp3s0

? (172.16.10.118) at d8:bb:c1:c5:6a:a4 [ether] on enp3s0

? (172.16.10.85) at d8:bb:c1:c5:cc:c3 [ether] on enp3s0

? (172.16.10.121) at d8:bb:c1:c5:cc:b5 [ether] on enp3s0

ii). hostname :- Displays the host name or computers name :

```
[codexam]$ hostname
localhost.localdomain
```

iii). ipconfig /all :-

This command displays detailed configuration information about your connection of network devices connected to our system

```
C:\User> ipconfig /all
```

Output :-

Ethernet adapter VMware VMnet8:

Connection-specific DNS Suffix:
Link-local IPV6 Address ....: fe80::dc61:d512:0
IPV4 Address ..... : 192.168.229.1
Subnet Mask ..... : 255.255.255.0

iv). nbtstat -a :- Resolves the problem with NetBIOS name resolution information over TCP/IP

C:\Users > nbtstat -a labpc1

Output :-

Local Area Connection:

Node IPAddress : [192.168.1.10] Scope Id: [ ]

NetBIOS Remote Machine Name Table

Name	Type	Status
labpc1	<01> Workgroup	Registered
labpc2	<01> workgroup	Registered
labpc3	<01> workgroup	Registered

v). nslookup :- Performs DNS both forward look up and reverse lookup to display the DNS name for IP and IP for DNS and operates on two modes interactive mode and non-interactive mode.

C:\User > nslookup google.com

Server : 192.168.1.1

Address : 192.168.1.1

C:\User > nslookup 8.8.8.8

vi). Pathping :- This is a Windows command which is a combined for ping and Tracert (Trace router) command which launches a 25 second test

```
C:\User>Pathping google.com
```

Tracing route to google.com [142.250.182.46]
over maximum 30 hops:

0 AshwanthanAchu [x.x.x.x]

1. x.x.x.x
2. 223.178.72.1
4. 116.119.36.00
5. 182.79.198.0
6. * * 142.250.169.206
7. 142.251.66.185
8. 142.250.62.66
9. * * *

vii). Ping :- To test connectivity through ICMP protocol. This command operates on both windows and Linux operating system.

Types :- i). Host name

ii). IP address

iii). FDNS naming

`C:\User> ping 8.8.8.8`

Puinging 8.8.8.8 with 32 bytes of data

Reply from 8.8.8.8 : bytes = 32 time = 6ms TTL = 119

Reply from 8.8.8.8 : bytes = 32 time = 7ms TTL = 119

Reply from 8.8.8.8 : bytes = 32 time = 4ms TTL = 119

Ping statistics for 8.8.8.8 :

Packets: Sent = 3, Recieved = 3, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 4ms, Maximum = 7ms, Average = 5ms

viii). **Route :-** This command shows the IP routing table primarily used in static routing.

`user $ ip route`

default via 169.254.1.254 dev ipvlan-eth0

192.254.0.0/16 via 169.254.1.254 dev ipvlan-eth

9x). ip :-

Syntax :-

ip <options> <object> <command>

*. To show an ip address assigned to an interface of your server :-

[root @ server]# ip address show

1: lo: <Loopback, up, lower_up> mtu 65536 qdisk noqueue state unknown group default qlen 1000

2: eth0: <Broadcast, Multicast, Up, Lower UP> mtu 15000 qdisc fq_codel state Unknown group default qlen 1000

*. Display the route taken for IP 8.8.8.8

[route @ server]# ip route get 8.8.8.8

8.8.8.8 via 192.168.229.2 dev eth0

Src 192.168.229.146 uid 1000 cache

*. Add a route to 192.168.1.0/24 via gateway 192.168.1.254:-

[route@server]# ip route add default via 192.168.1.254 dev eth0

x). ifconfig :- This command is used to troubleshooting and configuring the network

[root@server]# ifconfig

inet 192.168.229.116 netmask 255.255.255.0 broadcast 192.168.229.255

inet6 fe80::20c:29ff:fe31:6f6e prefixlen 64 scopeid 0x20<link>

xi). mtr :- ping + tracert command in Bi linux used to troubleshoot the network setting in linux environment.

[route@server]# mtr google.com

Hostname	Loss	Snt	Last	Avg	Best	Worst
???	0.0%	127	0	0	0	3
-gateway	0.0%	127	15	20	30	5

xii. tcpdump :-

This command helps to capture and display packets shared via internet

[root@server]# tcpdump -D

1. eth0 [Up, Running, Connected]
2. any (Pseudo-device that capture on all interface) [Up, Running]
3. lo [Up, Running, Loopback]

Configuring ethernet connected by using nmcli :-

Procedure :-

i. List Network Manager

#nmcli connection show

ii. #nmcli connection add con-name <connection-name> ifname <device-name> type ethernet

iii. Rename profile :-

#nmcli connection modify "<new_name>"

iv. Display current settings

#nmcli connection show

v). Configure the IPV4 settings :-

* # nmcli connection modify "<name>" ipv4. method auto

vi). Configure IPV6 settings :-

nmcli connection modify "<name>" ipv6. method auto

vii). Activate the profile :-

nmcli connection up internal-LAN

* # ip address show eth0
* # ip route show (check default setting line)
* # ip -6 route show
* # cat /etc/resolv.conf
* # ping <host-ip-address>